```
=== HAMMING CODE - ERROR DETECTION AND CORRECTION ===

Enter number of 4-bit data blocks in the stream: 2

--- Block 1 ---
Enter 4 data bits (space separated, e.g. 1 0 1 1): 1 0 1 1
Original Data Bits      : 1011
Encoded Hamming Code (7 bits): 0110011
Enter bit position to introduce error (1-7, 0 = no error): 3
Transmitted Code (with error induced at pos 3): 0100011
Received Code           : 0100011
Checking for errors...
   Error detected at bit position: 3 -> Correcting...
Corrected Code          : 0110011
Recovered Data Bits     : 1011

--- Block 2 ---
Enter 4 data bits (space separated, e.g. 1 0 1 1): 0 1 1 0
Original Data Bits      : 0110
Encoded Hamming Code (7 bits): 1100110
Enter bit position to introduce error (1-7, 0 = no error): 6
Transmitted Code (with error induced at pos 6): 1100100
Received Code           : 1100100
Checking for errors...
   Error detected at bit position: 6 -> Correcting...
Corrected Code          : 1100110
Recovered Data Bits     : 0110
```

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=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send     : 8
Enter sender's window size           : 4
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss)         : 5

-----------------------------------------------
Sender Window : [Frame 0 - Frame 3]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node A --> Sending Frame 3
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node B --> Received Frame 3, sending ACK 3
Node A --> ACK 3 received. Sliding window forward.

-----------------------------------------------
Sender Window : [Frame 4 - Frame 7]
Node A --> Sending Frame 4
Node A --> Sending Frame 5
        (Frame 5 LOST in transit!)
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Frame 5 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending window starting from Frame 5.

-----------------------------------------------
Sender Window : [Frame 4 - Frame 7]
Node A --> Sending Frame 5
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node A --> ACK 7 received. Sliding window forward.

-----------------------------------------------
All 8 frames successfully transmitted from Node A to Node B.
```